Justice Vidal

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SUMMARY

Software Engineer with machine learning and full-stack experience and a strong foundation in debugging and deployment. Demonstrated leadership and AWS-based cloud development through work experience. Applied ML and embedded systems knowledge from MIT coursework in fine-tuning LLMs using PyTorch and thesis on distributed TinyML inference using TensorFlow.

SKILLS

- **Programming & Development**: Python, PyTorch, TensorFlow, C++, React JS, CSS, Django, Typescript, SQLite, RESTful APIs, Object-oriented Design
- **Tools & Technologies**: AWS, PowerApps, Word, GitHub Desktop, LTspice, PSOC Creator, Spectre Circuit Simulator
- **Hardware**: PSOC 5 Microcontroller, STM32, ESP32, Soldering, Familiarity with core computer components

EXPERIENCE

MIT Digital Humanities Lab **Sep 2019 - Aug 2024**
Research Assistant/Lead Full Stack Developer *Cambridge, MA*

- Directed a team of 9 undergraduate students to develop Indian election data visualizations using object-oriented design principles and deployed the website on AWS, ensuring adherence to coding standards.
- Managed 10 students across 3 teams to create research articles for the 'This Was Paris in 1970' website, demonstrating clear design decisions and the site's utility to historians.
- Collaborated with 3 peers to evaluate object detection machine learning models for "This was Paris in 1970" online photo archive, presenting analytical findings at the 2021 Connecticut Digital Humanities Conference.

Keolis Commuter Services **Jun 2021 - Jun 2023**
Full Stack Developer Intern *Boston, MA*

- Reported to the Director of Service Delivery Innovation while designing and implementing new internal applications, integrating full-stack development practices and maintaining source control management protocols.
- Developed a system for storing qualitative employee performance data using feedback from the Operations Integration Manager, which involved coding, debugging, and validating the internal app for recording Train Conductor feedback.
- Collaborated with the Operations Planning team to rebuild the certification management website using modern coding practices and Power Apps, ensuring effective code reviews and final product presentation to key stakeholders including MBTA and the Keolis COO.

EDUCATION

Massachusetts Institute of Technology (MIT) **Feb 2025**
Masters, Electrical Engineering and Computer Science – Concentration: Artificial Intelligence

- **Coursework:** TinyML & Efficient Deep Learning Computing, Analysis and Design of Digital Integrated Circuits, Large-scale Symbolic Systems, Hardware Architecture for Deep Learning

Massachusetts Institute of Technology (MIT) **Jun 2023**
Bachelor of Science, Electrical Engineering and Computer Science, Physics – Minor: Philosophy

- **Coursework:** Computation Structures, Interconnected Embedded Systems, Microcomputer Project Laboratory, Modeling with Machine Learning: from Algorithms to Applications, Modern Machine Learning: Simple Methods that Work, Software Construction

PROJECTS

Scalable Embedded Tiny Machine Learning (SETML) **Aug 2023 - Feb 2025**
MEng Thesis *Cambridge, MA*

- Designed TensorFlow infrastructure in Python to split ML model into segments that could fit on resource constrained devices; Devices holding different parts of the model then communicate to complete inference on an input
- Implemented UART based protocol that left space for 2928 more parameters than standard Arduino I²C Wire interface while being more robust than base UART implementation

Mini-GPTs: Efficient Large Language Models through Contextual Pruning **Dec 2023**
Course: TinyML & Efficient Deep Learning Computing *Cambridge, MA*

- Designed and implemented the project's fine-tuning and multiple-choice based evaluation infrastructure in PyTorch
- Found that Microsoft Phi-1.5 translation performance could be recovered in 5 epochs after pruning to 69.069% of original size

LEADERSHIP

Student Information Processing Board (SIPB) **Sep 2019 - Dec 2024**
President/System Administrator

- Lead the Hardware Operations teams in restructuring the club's server room hosting campus-wide services; Dedicated production racks with UPSs and a separate development rack implemented to enhance service reliability and uptime.
- Served as President for 1 term and led weekly club meetings, fostering collaboration among members to coordinate their services